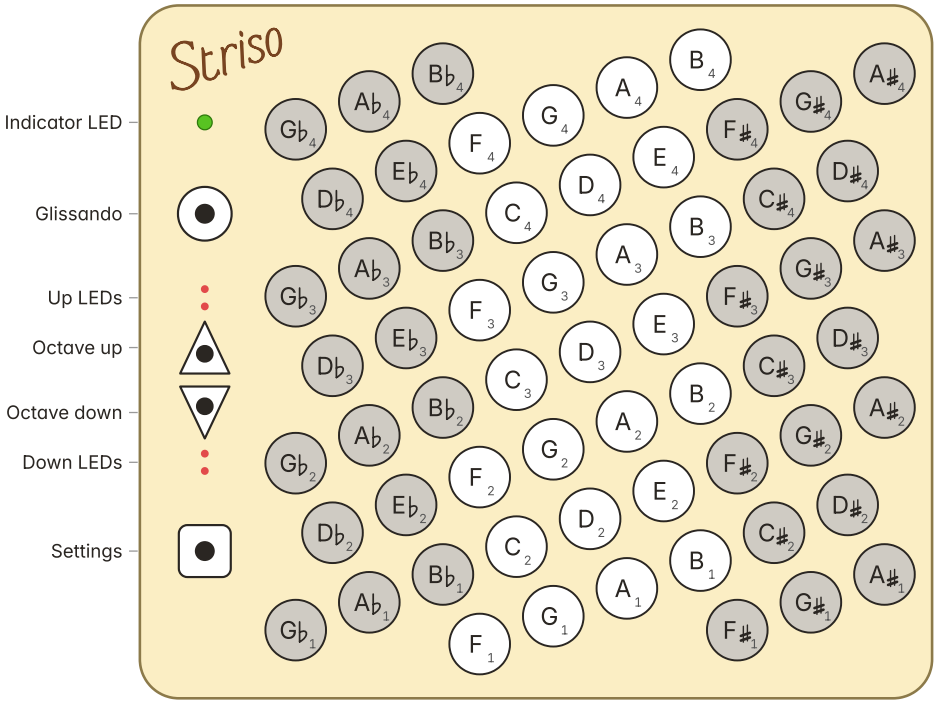
Striso

unofficial fork

Quick start guide

Striso board with the expressive synth fork firmware

The Striso board is a standalone instrument: plug in headphones or a speaker and play. With this firmware it carries nine sound engines (bowed strings, woodwinds, brass, a physical trumpet, an accordion, electric pianos, tuned percussion, a virtual analog synth and the original strisy sound), plays over USB as a sound card, and still works as an MPE MIDI controller.



61 keys, 4 buttons. Every key senses pressure, strike velocity and tilt in two directions. The isomorphic layout gives every chord and scale one shape in every key: a fifth is one column to the right, an octave is straight up. Octave numbers are the ones the configuration editor uses; D_3 in the middle sounds the D above middle C.

Thank you, Piers

The Striso is one of the most expressive instruments around. Every key senses pressure, strike velocity and tilt in two directions, and pressing one feels like touching a string, not pushing a button. The Striso is designed and built by **Piers Titus van der Torren**, along with its hardware, firmware, bootloader, configuration editor and this guide's original text and drawings: all of it thoughtful, careful work.

Piers released the firmware as **open source** under the GPL. That is the only reason this fork can exist: it builds directly on his firmware, his key sensing, his MIDI/MPE implementation and his bootloader. Thank you for making something this good and for sharing it. If you like what you hear, the instrument deserves the credit: **striso.com**, **forum.striso.com**.

About this fork

This is an **unofficial fork**, not supported by Striso. Its additions were vibe coded: written with an AI coding assistant (Claude Code), directed by a player saying what sounded wrong. They were tested on a real Striso board, and they work really well. It is still a fork made for fun, so keep a backup of your current firmware (copy `CURRENT.UF2`, see *Updating firmware*).

Everything the official firmware does still works: MPE and normal MIDI, tunings, presets, the configuration editor and the UF2 bootloader. **Striso Live**, the desktop app on page 7, is an optional companion; nothing needs it.

Getting started

Standalone: plug the USB cable into your Striso board and a power source, such as a phone charger, computer or power bank. Plug a headphone or speaker into the headphone / line out mini-jack and start playing.

MIDI controller: connect your Striso board to a synthesizer, computer or iPad with the USB cable or a TRS-MIDI type A adapter. If the synthesizer doesn't support MPE, change the MIDI mode to 'Normal' (see the settings card).

Buttons: the **octave up** and **down** buttons shift the sound up to two octaves up or down. The **glissando** button makes a smooth transition from one note to another. The square **settings** button turns the keys into settings keys while it is held (page 4).

Visit the Striso forum (forum.striso.com) for questions and discussions on anything Striso related. For the fork itself, use the issues at github.com/aidin-foroughi/striso-control-firmware.

The sounds

The on-board synth has nine voice engines. Several are physical models that react to the Striso's touch the way the real instrument reacts to a player.

- **Bowed strings:** a waveguide string driven by a friction bow, with a modelled body. Pressure is the bow, tilt moves it from fingerboard to bridge, a hard strike gives a short bite; pizzicato included.
- **Woodwinds:** modelled reeds and flute (clarinets, saxophones, oboe, bassoon). Pressure is breath; a fast strike tongues the note.
- **Brass:** a lip-reed model with a physical bore, from trumpet to tuba, with harmon and plunger mutes.
- **Trumpet, accordion:** a physical trumpet; free reeds with registers and one bellows.
- **Electric pianos:** physically inspired tines and reeds (Rhodes, Wurlitzer, Pianet and more).
- **Tuned percussion:** modal synthesis of real partials (vibraphone, marimba, glockenspiel, bell, kalimba, wine glass); pressure after the strike rubs and sustains.
- **Virtual analog:** saws, sync, a ladder filter and expressive envelopes.
- **strisy:** the original Striso sound, the default at power on.

All of them play with **10-voice polyphony** through **stereo master effects**: chorus, delay, reverb and tremolo. Every voice responds to pressure, strike velocity, tilt and the pedal. Page 4 explains how to pick a voice and a preset on the board; the table lists them in the order the up/down LEDs count.

LED	Voice	Presets
○	**strisy** the original Striso sound	the original sound (no presets)
🟠	**VA** virtual analog synth	classic, pad, lead, bass, brass, keys, acid, strings
🔵	**modal** tuned percussion	vibraphone, marimba, glockenspiel, bell, kalimba, wine glass
🔴	**bowed** bowed strings	violin, viola, cello, bass, section, ponticello, flautando, sordino, pizzicato
🟢	**wind** woodwinds	clarinet, bass clarinet, soprano sax, alto sax, tenor sax, baritone sax, oboe, bassoon, flute
🟣	**EP** electric pianos	stage, suitcase, mk2, dyno, ballad, bark, neosoul, fm80s, lofi, wurli, wurlibark, pianet, felt, celesta
🟡	**brass** brass	trumpet, flugelhorn, trombone, horn, euphonium, tuba, harmon, plunger, section
🩵	**trumpet** a physical trumpet	trumpet
🩷	**accordion** free reeds and bellows	clarinet, bassoon, piccolo, violin, musette, organ, master, bandoneon, harmonium

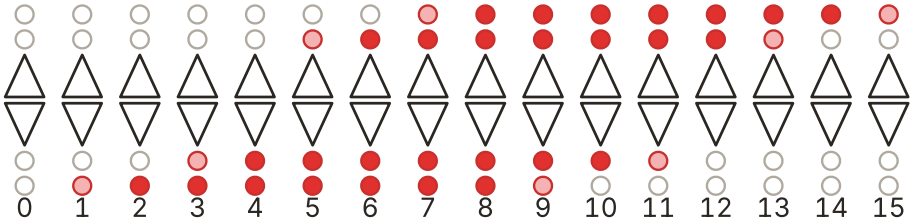
Striso Live (page 7) can shape every parameter of these engines and save your own sounds.

Settings

While the **settings** button is held, the keys adjust settings like volume or MIDI mode instead of playing. The card on page 5 shows which key does what.

Switch keys select something: a MIDI mode, a tuning, a preset, a voice. They act as soon as they are pressed, and the indicator LED lights in the matching colour.

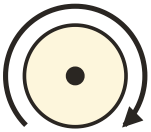
Dial keys (dashed ring on the card) hold a value: volume, a sensitivity, a voice's preset. Pressing one shows its value on the up/down LEDs, from 0 (all off) to 15:



Change the value by nudging the key up or down. For larger changes, make a circular motion with the key: clockwise for up, anticlockwise for down. Holding a nudge repeats it.



nudge up or down: one step



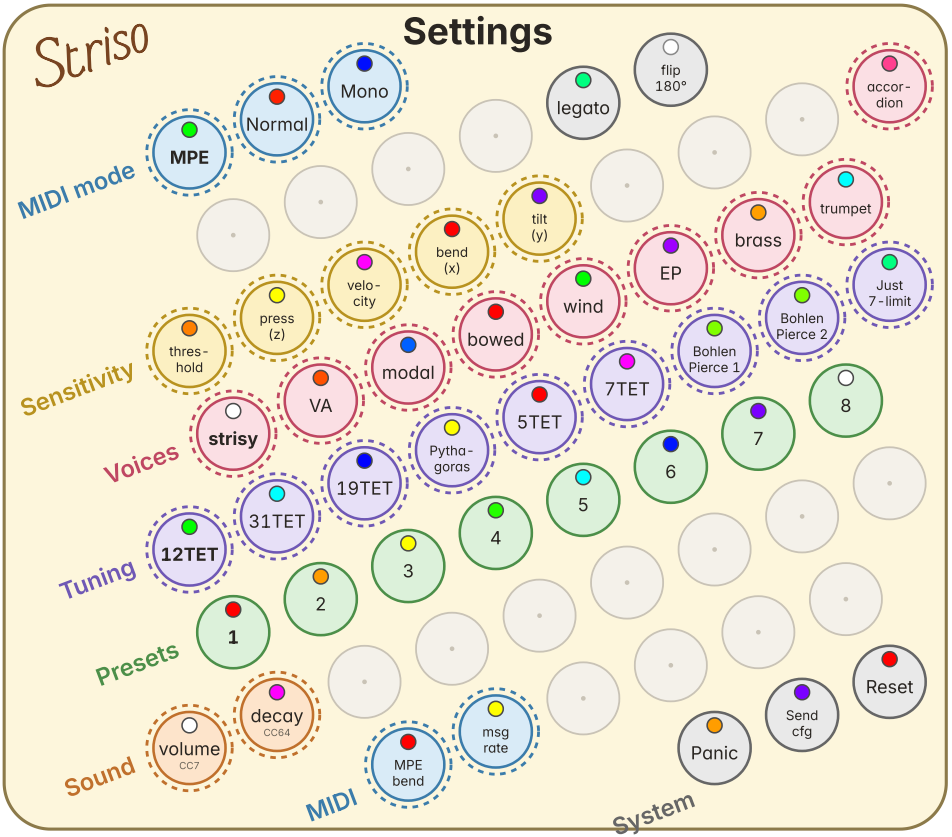
circle: clockwise up, anticlockwise down

> Settings changed on the board itself are not saved and return to their defaults when the Striso board is powered up again. Striso Live can save touch settings to the board.

Voices and presets

1. Hold the square **settings** button.
2. Press the key of a voice (the Voices row on the card). The indicator LED lights in the voice's colour and the voice plays with the preset it had last.
3. For another preset, keep the key pressed and **nudge it up or down**, or **circle** it. The up/down LEDs show the preset number, 1 being the first in the table on page 3. Each voice remembers its last preset until power off.

The **strisy** key has no presets and shows no number.



What the dials set. MPE: the number of MPE voices (1 to 15). Normal, Mono: the MIDI channel. Threshold: how light a touch starts a note. Press, velocity, bend, tilt: how strongly the touch counts. Voice keys: the preset. Tuning keys: a fine tuning offset, also for 12TET. Volume: 0 to 127. Decay: the sustain and decay time. MPE bend: 12, 24, 48 or 96 semitones. Msg rate: the MIDI message interval, 1 to 15 ms.

Switches. Presets 1 to 8 load stored settings. Tunings 1 to 8 are named as the factory sets them. Panic sends all notes off. Send cfg sends the MPE configuration again. **Flip 180°** turns the layout around. **Legato** switches mono legato on the internal synth on or off: one voice, a key pressed while another is held slurs to it, and its release returns to the held one. **Reset** restarts the board: hold settings, press Reset (red LED), then release settings. Blank keys do nothing yet.

Presets

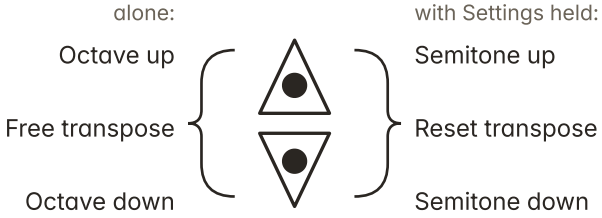
There is room for 8 presets to switch quickly between sets of settings. Presets are configured in the **configuration editor**: connect the Striso board to a computer with the USB cable while holding the settings button. The indicator LED blinks red and the Striso shows up as a USB drive; open the file `CONFIG.HTM` on it in your browser. A preset can also be chosen to load at power on.

Tuning and tuning systems

To match instruments that are not tuned to A440, nudge or circle the **12TET** key. Besides the common western 12-tone tuning, several other tuning systems are built in, like 31-tone equal temperament with its pure thirds, or just intonation where every interval to the C is pure. The tuning keys 1 to 8 load the tunings stored in the board; the names on the card are the factory ones. They can be customised in the configuration editor.

Transposing

The octave up and down buttons can also transpose by other intervals than octaves. Together with the settings button they transpose a semitone (a minor second in the current tuning) up or down. For a **free transpose**, hold both the up and the down button while playing a note, press the key where this note should sound, then release the first key: all keys move by that interval. Press the settings button together with the octave up and down buttons to reset any transpose.



Reset and factory settings

Restart: hold the settings button, press the bottom-right key (**Reset** on the card), then release the settings button. The board restarts with its saved settings.

Factory settings: hold the glissando button while plugging in. The board starts with its defaults instead of the saved power-on preset.

Update mode: plugging in with the settings button held starts the board as a USB drive (see *Updating firmware*). If you land there by accident, unplug and plug in again.

USB audio

Connected to a computer, the Striso also appears as a class-compliant USB sound card called **"Striso Synth"**: 48 kHz, stereo, 24 bit (16 bit is offered too), no driver needed on Windows or macOS. Record the internal synth straight into any DAW, or listen to it through the computer. The headphone jack keeps playing at the same time, and MIDI over the same USB cable works as before.

Striso Live, the optional companion app

Striso Live is a free desktop app for Windows and macOS. Nothing needs it; the Striso stays a standalone instrument. It makes the new sounds easy to explore. With it you can:

- turn every sound and effect parameter live while you play;
- switch presets and save your own sounds, and A/B them against the preset;
- see scales (66 of them), chord names and roman numerals on the board as you play;
- set the touch sensitivity and even out individual keys with a per-key equalizer, then save them to the board;
- hear the Striso through your computer and record it to WAV;
- play full screen in stage mode.

Download it from **github.com/aidin-foroughi/striso-live** (releases: a Windows installer, a Windows zip and a macOS disk image). The README there also has one-line installers for macOS and Windows.

Updating firmware

1. Plug in the Striso board with the **settings** (square) button held down. It attaches to the computer as a flash drive and the indicator LED blinks red.
2. Keep a backup first: copy `CURRENT.UF2` from the drive to your computer.
3. Copy the new `.uf2` file onto the drive. The board reboots by itself.

The version on the board is in `INFO_FW.TXT`. An update doesn't touch the bootloader or your saved settings. To go back, copy an official release from `github.com/striso/striso-control-firmware` or your backup the same way.

Fork firmware releases: **github.com/aidin-foroughi/striso-control-firmware**. Build it yourself with `tools/build.sh uf2 VOICE=all`; the README explains the toolchain.

Troubleshooting

The Striso board is bent, it wobbles on a flat surface.

→ Twist the sides gently with both hands. Try a couple of times until it doesn't wobble anymore.

There is a buzzing sound in the audio output.

→ There may be a ground loop if the USB cable is connected to the same device as the audio cable (possibly indirectly). Power the Striso board from an independent power source, try a better quality audio cable, or use a ground loop isolator. USB audio avoids the analog cable altogether.

The keys are dirty.

→ The keys can be cleaned with a damp cloth. Small particles around the keys can be removed with a toothpick.

There are random notes in the MIDI output.

→ Increase the sensitivity **threshold** a bit. The default is very sensitive, which works well with many synthesizers that react to velocity and aftertouch, but for some synthesizers and for notation software random notes are not acceptable.

The LED blinks pink.

→ The Striso board receives MIDI messages over USB. If the board behaves strangely, those messages may be the cause.

The sound is gone or very quiet.

→ Check the volume key and the voice keys on the settings card. Starting with factory settings (glissando button while plugging in) restores the defaults.

The computer doesn't show "Striso Synth".

→ Use a data USB cable (not a charge-only one) and a direct port. The audio device is part of the fork firmware; the official firmware has only MIDI.

Something else is wrong after the update.

→ Copy your backup or an official release back (see *Updating firmware*). Nothing in the fork changes the bootloader.

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